



IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

Faculté des Sciences et des Technologies

Laboratoire N° 8 _ Réseaux II

Niveau : Licence III

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Date : 19 Mai 2026

ÉTAPE 1 : Création de la VM (pfsense)

New Virtual Machine

Virtual machine name and operating system

VM Name

VM Folder

ISO Image

OS Edition

OS

OS Distribution

OS Version

☐ Proceed with Unattended Installation

> Set up unattended guest OS installation

> Specify virtual hardware

> Specify virtual hard disk

Base Memory

Number of CPUs

☐ Use EFI

Create a New Virtual Hard Disk

Hard Disk File Location and Size

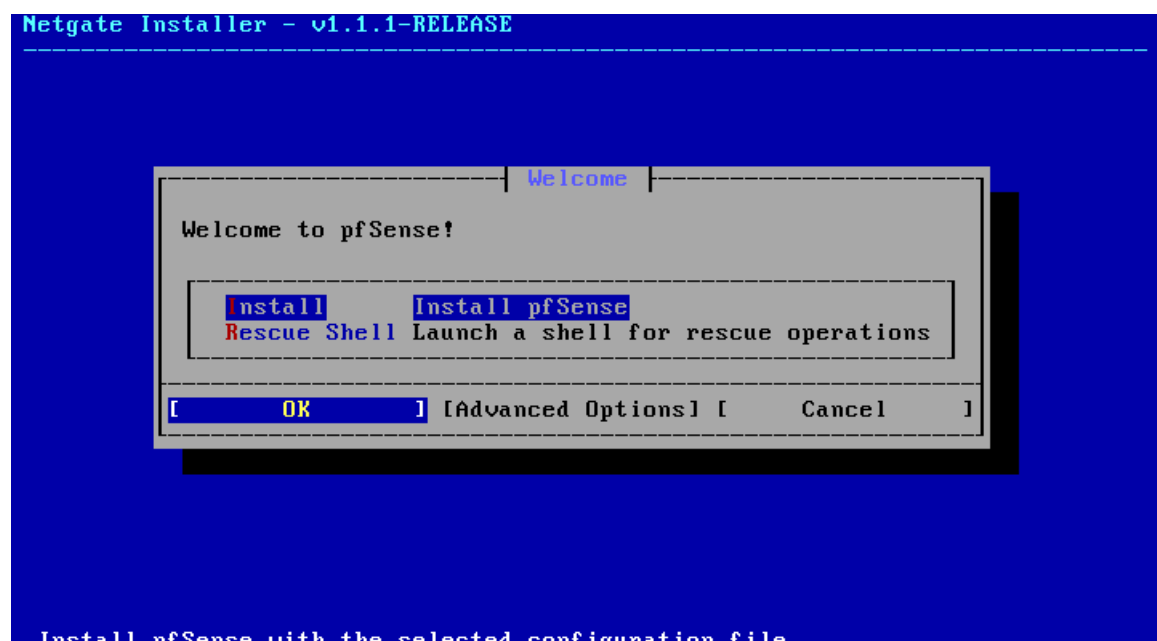
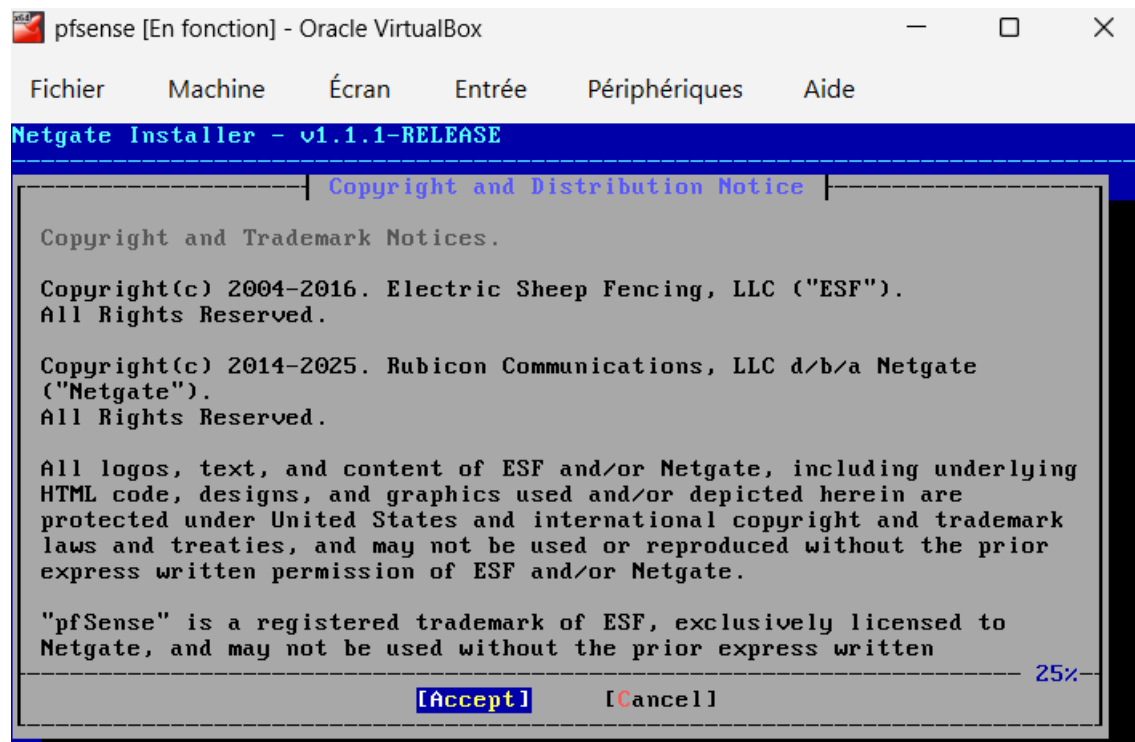
Disk Size

Hard Disk File Type and Format

☐ Pre-allocate Full Size

☐ Split Disk Into 2 GB Parts

ÉTAPE 2 : Installation de pfSense



Active Subscription Validation

This device does not have an active pfSense Plus subscription.

To purchase pfSense Plus:

Software Store: <https://www.netgate.com/purchase-plus>

NDI: a112 64d7 bf28 a86c 2b31

and select 'Retry Validation' once purchase is complete.

Or select 'Install CE' to install pfSense CE software.

[**Install CE**]

[**Retry Validation**]

[**Cancel**]

Software Version to Install

Select the version of pfSense CE to install.

0000-2_8_1 Current Stable Version (2.8.1)

0001-2_8_0 Previous Stable Version (2.8.0)

0002-2.7.2 Previous Stable Version (2.7.2)

[**OK**]

[**Cancel**]

Installation Details

```
Fetching meta.conf: . done
pkg-static: Failed to fetch https://pkg.pfsense.org/pfSense_v2_7_2_amd64-c
pkg-static: Failed to fetch https://pkg.pfsense.org/pfSense_v2_7_2_amd64-c
Fetching packagesite.pkg: . done
Processing entries: . done
pfSense-core repository update completed. 4 packages processed.
Updating pfSense repository catalogue...
Fetching meta.conf: . done
Fetching data.pkg: ..... done
Processing entries:
Processing entries..... done
pfSense repository update completed. 550 packages processed.
All repositories are up to date.
The following 1 package(s) will be affected (of 0 checked):

New packages to be INSTALLED:
  pkg: 1.20.8_3 [pfSense]

Number of packages to be installed: 1

The process will require 39 MiB more space.
10 MiB to be downloaded.
```

Netgate Installer - v1.1.1-RELEASE

Complete

Installation of pfSense complete! Would you like to reboot into the installed system now?

[Reboot]

[Shell]

pfSense 2.7.2-RELEASE amd64 20240304-1953
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: a11264d7bf28a86c2b31

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
 v6/DHCP6: fd17:625c:f037:2:a00:27ff:feed:afc/64
LAN (lan) -> em1 -> v4: 192.168.1.1/24

- | | |
|-----------------------------------|----------------------------------|
| 0) Logout (SSH only) | 9) pfTop |
| 1) Assign Interfaces | 10) Filter Logs |
| 2) Set interface(s) IP address | 11) Restart webConfigurator |
| 3) Reset webConfigurator password | 12) PHP shell + pfSense tools |
| 4) Reset to factory defaults | 13) Update from console |
| 5) Reboot system | 14) Enable Secure Shell (sshd) |
| 6) Halt system | 15) Restore recent configuration |
| 7) Ping host | 16) Restart PHP-FPM |
| 8) Shell | |

Enter an option: █

```

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.2.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) y

```

```

> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) y

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to LAN...
Reloading filter...
Reloading routing configuration...
DHCPD...

The IPv4 LAN address has been set to 192.168.2.1/24

The IPv6 LAN address has been set to dhcp6

Press <ENTER> to continue.

```

```

The IPv6 LAN address has been set to dhcp6

Press <ENTER> to continue.
VirtualBox Virtual Machine - Netgate Device ID: a11264d7bf28a86c2b31

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

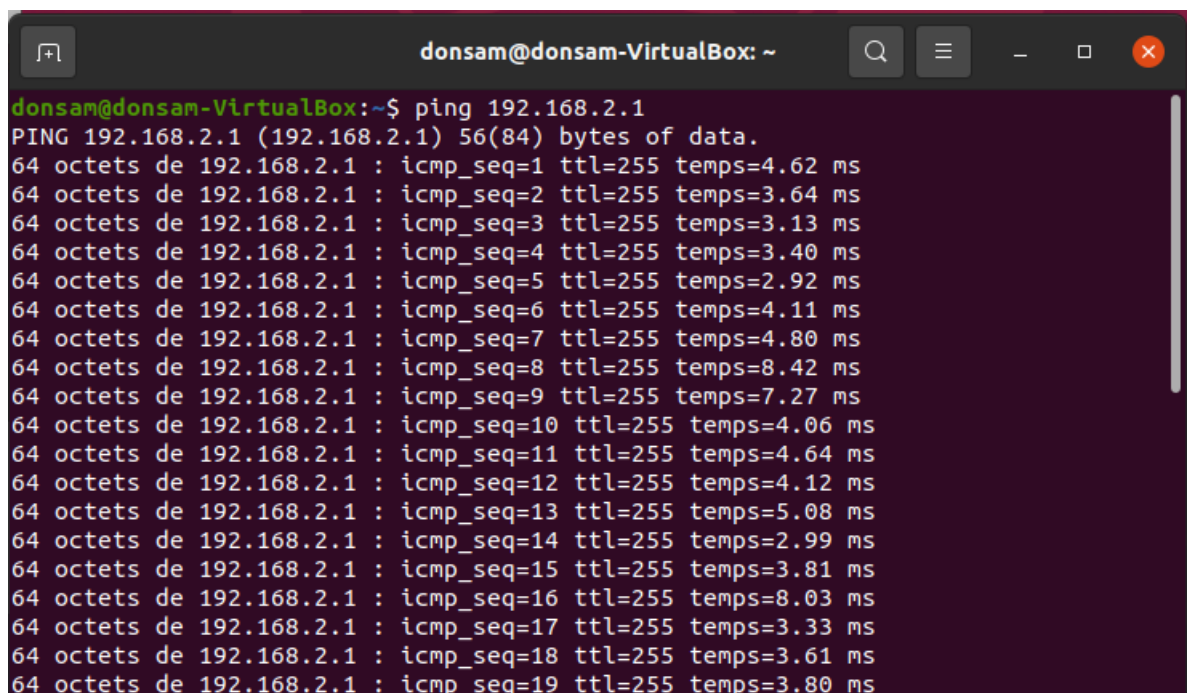
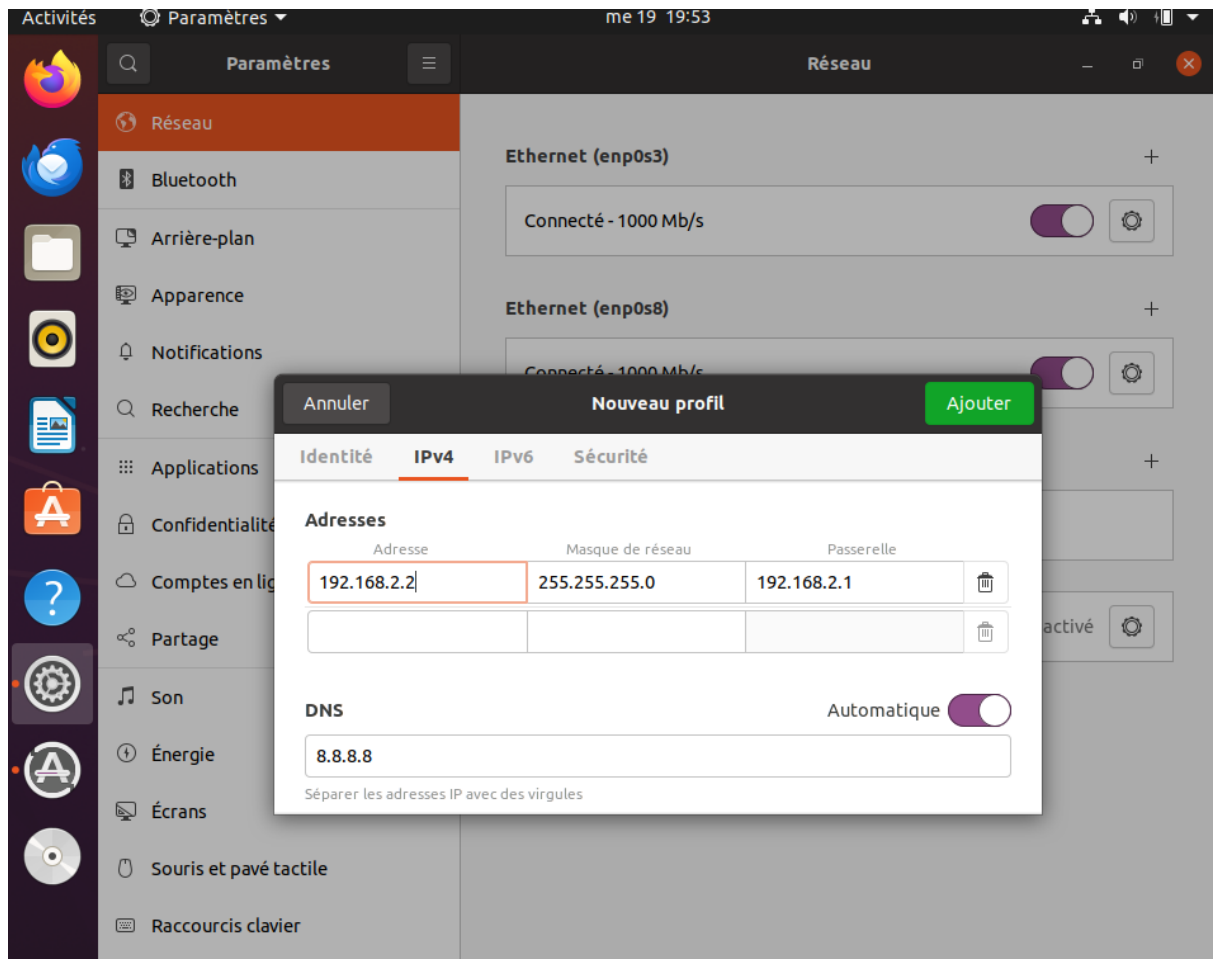
WAN (wan)      -> em0      -> v4/DHCP4: 10.0.2.15/24
                                   v6/DHCP6: fd17:625c:f037:2:a00:27ff:feed:afc
4
LAN (lan)      -> em1      -> v4: 192.168.2.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

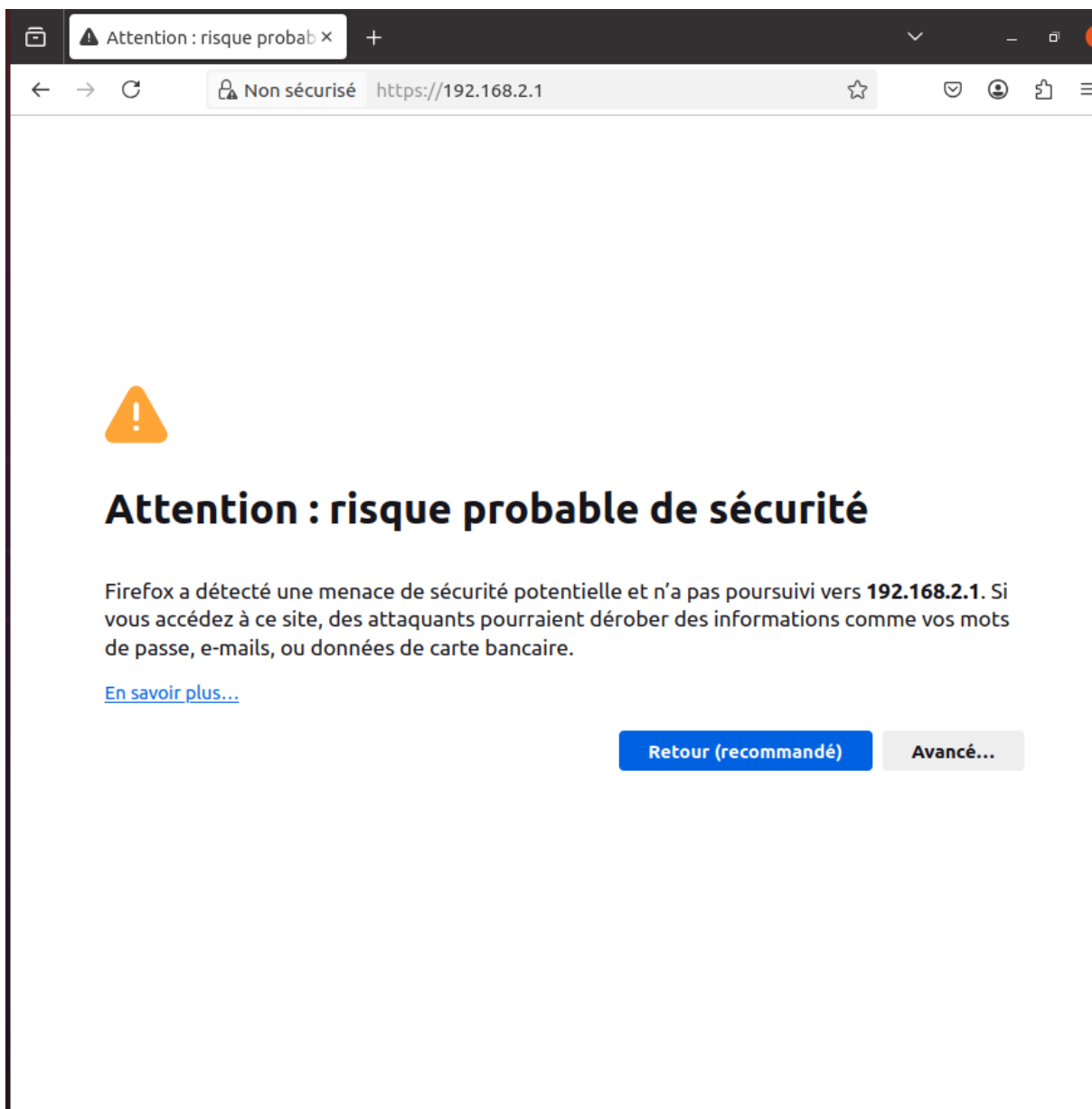
Enter an option:

```

Sur Ubuntu-VM



✓ Accès à l'interface Web





Connectez-vous à pfSense

CONNECTEZ-VOUS

admin

.....|

CONNECTEZ-VOUS

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Wizard / [pfSense Setup](#) /

Step

pfSense Setup

Welcome to pfSense® software!

This wizard will provide guidance through the initial configuration of pfSense.

The wizard may be stopped at any time by clicking the logo image at the top of the screen.

pfSense® software is developed and maintained by Netgate®

[Learn more](#)

[» Next](#)

❖ CONFIGURATION DE BASE pfSense

← → ↻ https://192.168.2.1/wizard.php?xml=setup_wizard.xml 📄 🔍 ☆ 📧 👤 📁 ⋮

Step 2 of 5

General Information

On this screen the general pfSense parameters will be set.

Hostname
Name of the firewall host, without domain part.
Examples: pfsense, firewall, edgefw

Domain
Domain name for the firewall.
Examples: home.arpa, example.com

Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The 'local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses 'local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.

The default behavior of the DNS Resolver will ignore manually configured DNS servers for client queries and query root DNS servers directly. To use the manually configured DNS servers below for client queries, visit Services > DNS Resolver and enable DNS Query Forwarding after completing the wizard.

Primary DNS Server

Secondary DNS Server

Override DNS ☒
Allow DNS servers to be overridden by DHCP/PPP on WAN

[» Next](#)

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Wizard / [pfSense Setup](#) / Time Server Information

Step 3 of 9

Time Server Information

Please enter the time, date and time zone.

Time server
hostname

2.pfsense.pool.ntp.org

Enter the hostname (FQDN) of the time server.

Timezone

Canada/Eastern

[»»](#) Next

Step 5 of 9

Configure LAN Interface

On this screen the Local Area Network information will be configured.

LAN IP
Address

192.168.2.1

Type dhcp if this interface uses DHCP to obtain its IP address.

Subnet Mask

24

[»»](#) Next

**Wizard completed.****Congratulations! pfSense is now configured.**

We recommend that you check to see if there are any software updates available. Keeping your software up to date is one of the most important things you can do to maintain the security of your network.

[Check for updates](#)**Remember, we're here to help.**

[Click here](#) to learn about Netgate 24/7/365 support services.

User survey

Please help all the people involved in improving and expanding pfSense software by taking a moment to answer this short survey (all answers are anonymous)

[Anonymous User Survey](#)

Useful resources.

- Learn more about Netgate's product line, services, and pfSense software from our [website](#)
- To learn about Netgate appliances and other offers, [visit our store](#)
- Become part of the pfSense community. Visit our [forum](#)
- Subscribe to our [newsletter](#) for ongoing product information, software announcements and special offers.

[Finish](#)

pfSense-lab.domain.local x

+

https://192.168.2.1

🔍 ⭐ 🛡️ 👤 📄 ☰

 pfSense
COMMUNITY EDITION

☰

Status / Dashboard

+

?

System Information

⚙️ - ✕

Name	pfSense-lab.domain.local
User	admin@192.168.2.2 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: a11264d7bf28a86c2b31
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006
Version	2.7.2-RELEASE (amd64) built on Mon Mar 4 14:53:00 EST 2024 FreeBSD 14.0-CURRENT The system is on the latest version. Version information updated at Tue May 19 20:55:00 EDT 2026 ↻
CPU Type	12th Gen Intel(R) Core(TM) i5-12450H 2 CPUs: 1 package(s) x 2 cache groups x 1 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	00 Hour 48 Minutes 57 Seconds
Current date/time	Tue May 19 21:42:40 EDT 2026
DNS server(s)	127.0.0.1 192.168.2.1

CONFIGURATION FIREWALL

➤ Bloquer l'accès à YouTube par IP sur pfSense

pfSense-lab.domain.local x

https://192.168.2.1/firewall_aliases_edit.php?tab=ip

pfSense
COMMUNITY EDITION

Firewall / Aliases / Edit

Properties

Name YOUTUBE_BLOCK
The name of the alias may only consist of the characters "a-z, A-Z, 0-9 and _".

Description Bloquer acces Youtube
A description may be entered here for administrative reference (not parsed).

Type Network(s)

Network(s)

Hint Networks are specified in CIDR format. Select the CIDR mask that pertains to each entry. /32 specifies a single IPv4 host, /128 specifies a single IPv6 host, /24 specifies 255.255.255.0, /64 specifies a normal IPv6 network, etc. Hostnames (FQDNs) may also be specified, using a /32 mask for IPv4 or /128 for IPv6. An IP range such as 192.168.1.1-192.168.1.254 may also be entered and a list of CIDR networks will be derived to fill the range.

Network or FQDN	142.250.0.0 / 16	Description	Delete
	172.217.0.0 / 16	Description	Delete
	216.58.0.0 / 16	Description	Delete

Save + Add Network

➤ Bloquer l'accès à FaceBook par IP sur pfSense

 COMMUNITY EDITION



Firewall / Aliases / Edit 

Properties

Name	FACEBOOK_BLOCK
The name of the alias may only consist of the characters "a-z, A-Z, 0-9 and _".	
Description	Bloquer acces Facebook
A description may be entered here for administrative reference (not parsed).	
Type	Network(s) 

Network(s)

Hint Networks are specified in CIDR format. Select the CIDR mask that pertains to each entry. /32 specifies a single IPv4 host, /128 specifies a single IPv6 host, /24 specifies 255.255.255.0, /64 specifies a normal IPv6 network, etc. Hostnames (FQDNs) may also be specified, using a /32 mask for IPv4 or /128 for IPv6. An IP range such as 192.168.1.1-192.168.1.254 may also be entered and a list of CIDR networks will be derived to fill the range.

Network or FQDN	157.240.0.0 / 16 	Entry added Tue, 19 May 2026 22:00	 Delete
	31.13.24.0 / 21 	Entry added Tue, 19 May 2026 22:00	 Delete
	31.13.64.0 / 18 	Entry added Tue, 19 May 2026 22:00	 Delete
	66.220.144.0 / 20 	Entry added Tue, 19 May 2026 22:00	 Delete

 Save  Export to file  Add Network

➤ Bloquer l'accès à Tiktok par IP sur pfSense

Firewall / Aliases / Edit ?

Properties

Name

TIKTOK_BLOCK

The name of the alias may only consist of the characters "a-z, A-Z, 0-9 and _".

Description

Bloquer acces Tiktok

A description may be entered here for administrative reference (not parsed).

Type

Network(s)

▼

Network(s)

Hint

Networks are specified in CIDR format. Select the CIDR mask that pertains to each entry. /32 specifies a single IPv4 host, /128 specifies a single IPv6 host, /24 specifies 255.255.255.0, /64 specifies a normal IPv6 network, etc. Hostnames (FQDNs) may also be specified, using a /32 mask for IPv4 or /128 for IPv6. An IP range such as 192.168.1.1-192.168.1.254 may also be entered and a list of CIDR networks will be derived to fill the range.

Network or FQDN

103.250.140.0 / 22 ▼

Description

Delete

162.159.138.0 / 24 ▼

Description

Delete

172.64.140.0 / 24 ▼

Description

Delete

Save

+ Add Network

IP Ports URLs All

Firewall Aliases IP

Name	Type	Values	Description	Actions
FACEBOOK_BLOCK	Network(s)	157.240.0.0/16, 31.13.24.0/21, 31.13.64.0/18, 66.220.144.0/20	Bloquer acces Facebook	  
TIKTOK_BLOCK	Network(s)	103.250.140.0/22, 162.159.138.0/24, 172.64.140.0/24	Bloquer acces Tiktok	  
YOUTUBE_BLOCK	Network(s)	142.250.0.0/16, 172.217.0.0/16, 216.58.0.0/16	Bloquer acces Youtube	  

+ Add

i

Floating

WAN

LAN

➤ Création du Schedule (Planning horaire)

Month

May_26

Date

May_2026						
Mon	Tue	Wed	Thu	Fri	Sat	Sun
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Click individual date to select that date only. Click the appropriate weekday Header to select all occurrences of that weekday.

Time

0

00

23

59

Start Hrs
Start Mins
Stop Hrs
Stop Mins

Select the time range for the day(s) selected on the Month(s) above. A full day is 0:00-23:59.

Time range description

A description may be entered here for administrative reference (not parsed).

+ Add Time

↶ Clear selection

Configured Ranges

Mon - Fri	8:00	18:00		Delete
Day(s)	Start time	Stop time	Description	

COMMUNITY EDITION

Firewall / Schedules

?

Schedules

Name	Range: Date / Times / Name	Description	Actions
WORKING_HOURS	Mon - Fri / 8:00-18:00 /	Acces Internet autorise pendant les heures de travail	

Indicates that the schedule is currently active.

+ Add

➤ Règle Firewall pour le Schedule

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	1/2.13 MiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none		TIKTOK_BLOCK	
☐	0/4 KiB	IPv4 *	LAN subnets	*	*	*	*	none		FACEBOOK_BLOCK	
☐	0/0 B	IPv4 *	LAN subnets	*	*	*	*	none	WORKING_HOURS	Bloquer internet hors horaires	
☐	0/2 KiB	IPv4 *	LAN subnets	*	YOUTUBE_BLOCK	*	*	none		YOUTUBE_BLOCK	
☐	0/40.51 MiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Add
 Add
 Delete
 Toggle
 Copy
 Save
 Separator

Test

✓ TEST de Connectivité de base

```
donsam@donsam-VirtualBox:~$ ping -c 3 192.168.2.1
PING 192.168.2.1 (192.168.2.1) 56(84) bytes of data.
64 octets de 192.168.2.1 : icmp_seq=1 ttl=64 temps=3.06 ms
64 octets de 192.168.2.1 : icmp_seq=2 ttl=64 temps=3.91 ms
64 octets de 192.168.2.1 : icmp_seq=3 ttl=64 temps=1.76 ms

--- statistiques ping 192.168.2.1 ---
3 paquets transmis, 3 reçus, 0 % paquets perdus, temps 2003 ms
rtt min/moy/max/mdev = 1,758/2,907/3,905/0,882 ms
```

✓ TEST de Blocage YouTube par IP

```
donsam@donsam-VirtualBox:~$ ping -c 3 142.250.0.1
PING 142.250.0.1 (142.250.0.1) 56(84) bytes of data.

--- statistiques ping 142.250.0.1 ---
3 paquets transmis, 0 reçus, 100 % paquets perdus, temps 2051 ms

donsam@donsam-VirtualBox:~$ ping -c 3 172.217.0.1
PING 172.217.0.1 (172.217.0.1) 56(84) bytes of data.

--- statistiques ping 172.217.0.1 ---
3 paquets transmis, 0 reçus, 100 % paquets perdus, temps 2036 ms

donsam@donsam-VirtualBox:~$ ping -c 3 216.58.0.1
PING 216.58.0.1 (216.58.0.1) 56(84) bytes of data.

--- statistiques ping 216.58.0.1 ---
3 paquets transmis, 0 reçus, 100 % paquets perdus, temps 2050 ms

donsam@donsam-VirtualBox:~$
```

✓ TEST de vérification logs firewall

 COMMUNITY EDITION

Status / System Logs / Firewall / Normal View

System Firewall DHCP Authentication IPsec PPP PPPoE/L2TP Server OpenVPN NTP

Packages Settings

Normal View Dynamic View Summary View

Last 30 Firewall Log Entries. (Maximum 500)

Action	Time	Interface	Rule	Source	Destination
✗	May 20 02:14:35	LAN	 YOUTUBE_BLOCK (1779242496)	 192.168.2.2	 142.250.0.1
✗	May 20 02:14:36	LAN	 YOUTUBE_BLOCK (1779242496)	 192.168.2.2	 142.250.0.1
✗	May 20 02:14:37	LAN	 YOUTUBE_BLOCK (1779242496)	 192.168.2.2	 142.250.0.1
✗	May 20 02:15:06	LAN	 YOUTUBE_BLOCK (1779242496)	 192.168.2.2	 172.217.0.1
✗	May 20 02:15:07	LAN	 YOUTUBE_BLOCK (1779242496)	 192.168.2.2	 172.217.0.1
✗	May 20 02:15:08	LAN		 192.168.2.2	 172.217.0.1

pfSense Firewall Log Entries showing blocked connections to YouTube IP addresses (142.250.0.1 and 172.217.0.1) from the LAN interface. The rule used is YOUTUBE_BLOCK (1779242496).

Questions

1. Quel est le rôle de pfSense ?

R - pfSense est un pare-feu open source basé sur FreeBSD. Il sécurise le réseau interne, gère le trafic Internet, applique des règles de filtrage, fait du NAT et permet le contrôle des accès par horaires.

2. Différence WAN et LAN ?

- **WAN** (Wide Area Network) : interface connectée à Internet (côté externe)
- **LAN** (Local Area Network) : interface connectée au réseau interne (côté machines clientes)

2. Pourquoi utiliser NAT automatique ?

R - Le NAT (Network Address Translation) permet à toutes les machines du réseau interne (192.168.2.x) de partager une seule IP publique pour accéder à Internet. Sans NAT, les machines internes ne pourraient pas communiquer avec Internet.

3. Comment fonctionne une règle firewall ?

R - pfSense lit les règles de haut en bas et applique la première règle qui correspond au trafic. Chaque règle définit une Action (PASS/BLOCK), une Interface, un protocole, une source et une destination. Si aucune règle ne correspond, le trafic est bloqué par défaut.

4. Pourquoi VirtualBox est utilisé pour les tests ?

R - VirtualBox permet de créer un environnement réseau virtualisé complet sans matériel physique. On peut simuler plusieurs machines (pfSense, clients) interconnectées de manière isolée et sécurisée.

5. Quel est le rôle de pfSense dans ce réseau ?

R - Dans le réseau de l'IUS, pfSense est le pare-feu central qui contrôle tout le trafic entre le réseau du campus et Internet. Il filtre les accès, bloque les sites non pédagogiques, applique des horaires d'accès et journalise toutes les connexions.

6. Pourquoi utiliser un Schedule ?

R - Un Schedule permet d'automatiser l'activation et la désactivation des règles firewall selon des plages horaires définies. Cela évite une intervention manuelle et permet de gérer l'accès Internet selon les besoins (heures de cours, heures de bureau).

8. Différence entre PASS et BLOCK dans les logs ?

- **PASS** : le trafic est autorisé à passer par le firewall
- **BLOCK** : le trafic est bloqué et abandonné silencieusement sans notification à l'expéditeur

Conclusion

Ce TD nous a permis d'installer et configurer pfSense sur VirtualBox afin de sécuriser un réseau d'entreprise. Nous avons mis en place des règles de filtrage pour bloquer les sites non autorisés (YouTube, Facebook, TikTok) et un planning horaire pour contrôler l'accès à Internet. Cela démontre l'efficacité de pfSense comme solution de sécurité réseau open source accessible et facile à déployer.